

Group 1 Members: Craig Vande Ligt, Akshaya Ramanarayanan, Anca Rotaru, Bhavin Prajapati, Collin Potter Bonar, Rob Mifsud

Topic: Comparison of Random Forest, Support Vector, and K-Nearest Neighbors Classifiers on Sentinel-2 Satellite Imagery

Objective: Land cover classification using satellite imagery is a vital application of remote sensing and machine learning. Accurate identification of surface types, such as forests, sea/lakes, highways, and residential zones, allow governments and organizations to monitor environmental change, plan infrastructure, and assess agricultural conditions.

The goal of this project was to apply several optimized supervised classification models to Sentinel-2 satellite imagery and compare their performance against the target data. Comparisons between the models' accuracies, precision, recall, f1-scores, ROC, and AUC were made, while identifying the strengths and weaknesses in each approach. This involved analyzing the models' behaviour under different levels of complexity and regularization.

Data Source and Details:

<https://www.kaggle.com/datasets/apollo2506/eurosat-dataset?resource=download-directory&select=EuroSATallBands>

Notes:

1. The above dataset contains 13 spectral bands of data, while the below "Full" and "Reduced" datasets only contain 4 spectral bands of data (blue, green, red, and near-infrared).
2. Due to the extreme computation loads of the "Full" dataset (~113 billion pixels and 4 columns of data - one column per spectral band), a reduced stratified dataset was built. While this reduced dataset made modeling possible, it still took minutes to hours to run each model, depending on the laptop's capabilities, even with parallel processing being utilized.

Full dataset pixel count: 113,037,312 Full dataset size: 534.334 MB

Reduced dataset pixel count: 100,000 Reduced dataset size: 0.708 MB

Model Design:

Three non-linear classification models were employed, Random Forest (RF/RFC), Support Vector Machine (SVM/SVC), and k-Nearest Neighbor (kNN), to test how various model structures would behave. RF models are known for their robust ensemble of decision trees and ability to avoid overfitting if properly tuned. SVMs are known for their ability to work with linear and non-linear data structures using various kernels, possibly making it a good fit for natural land feature identification. Both RF and SVMs are less sensitive to overtraining and noise. kNN is a well-known algorithm used in remote sensing image classification for decades. It is well known for its simplicity and ability to handle non-parametric data.

Model Evaluation:

Table 1. Summary of models' key performance metrics. "Generalization" is determined by the gap between training and testing accuracy; A gap over 10% indicates significant overfitting. The top-performing model is bolded.

Model	Train Accuracy	Test Accuracy	Macro F1-Score	Weighted F1-Score	Macro AUC	Training Time (s)	Generalization
RFC Model A	86.85%	56.32%	0.5281	0.55	0.8831	24.60	Poor (Overfit)
RFC Model B	55.68%	53.55%	0.4934	0.51	0.8734	9.53	Good
Support Vector Classifier	52.33%	51.93%	0.4586	0.48	0.8337	2599.55	Good
K-Nearest Neighbors	99.91%	54.83%	0.5158	0.53	0.8476	4.53	Poor (Overfit)

- **Generalization:** RFC Model B and SVC generalized well; RFC Model A and kNN overfit significantly
- **Overall Performance:** RFC Model A had best scores, but overfit; RFC Model B offered the most balanced and trustworthy performance among well-generalized models; SVC had the lowest overall metrics
- **Computational Efficiency:** kNN was quickest, followed by RFC models. SVC was significantly slower
- **Class-specific Performance:** All models struggled with "River" and "Highway" classes, and to a lesser extent Herbaceous Vegetation. SeaLake and Forest were consistently well predicted among all models. Interestingly, the SVC model struggled significantly with the "River" class compared to all other models.
- **Suitability:** The well-generalized Random Forest Model B was deemed the most suitable due to its balanced performance, effective generalization, and efficiency. Overfitted models are unstable, and the Support Vector Machine model was impractical due to computation load, training time, and poor critical class performance.

Future Work:

The analysis suggests that while the current classification models show promise, there is significant room for improvement, particularly in addressing class imbalance and exploring more advanced modeling techniques.

- **Apply resampling techniques** like SMOTE (Synthetic Minority Over-sampling Technique) to generate synthetic data points for rare classes
- **Use boosting algorithms**, such as XGBoost and LightGBM, which are often superior to standard Gradient Boosted Trees and inherently handle class imbalance
- **Neural Networks** could be utilized, provided their architecture is appropriately designed
- **Further SVC optimization** could be done by exploring a wider range of kernel types
- Use **soft voting ensemble model** to determine the best metrics among existing classification models
- Evaluate use of an **extra boost random forest classification** model to enhance performance by strengthening the weakest portions of each tree via random selection